MIDTERM REPORT PBL

MACHINE LEARNING

UrbanHeat AI: Spatio-Temporal Prediction of Urban Heat Intensity Using Machine Learning

Group 4

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Executive Summary

UrbanHeat AI is a mobile-oriented project designed to support the exploration of land surface temperature (LST) patterns in Malang City, Indonesia. The project addresses the need to understand urban heat conditions spatially and to communicate temperature-related information in an accessible way. Its machine-learning component estimates LST in degrees Celsius from geographic, temporal, and environmental inputs.

Based on recent evaluations and technical feedback, the primary focus of development is currently centered on data preprocessing and feature engineering optimization. Previously, using raw data representations and features without direct linear correlation caused severe model instability, leading to poor or negative R^2 validation scores.

To solve these challenges, the team has finalized the following data strategies:

- Weekly Temporal Aggregation = from daily to weekly basis
- Cyclical Time Encoding: using Sine and Cosine functions to preserve natural seasonal continuity in regression models.
- Missing Data Imputation = Missing LST pixels caused by cloud cover are filled using location-wise mean imputation across matching spatial coordinates rather than being dropped
- Feature Enrichment & Selection = The model incorporates NDVI, NDBI, a composite interaction term ($NDVI \times NDBI$), and spatial coordinates (Longitude & Latitude). Before final model selection, all variables must undergo Pearson correlation (R) and R^2 screening to eliminate uninformative features.

Dataset

The initial dataset was constructed from raw MOD11A1 MODIS satellite observations covering Malang City, comprising daily records across 116 spatial points with approximately 150 observation dates. In this preliminary approach, the raw feature set included spatial coordinates (longitude and latitude), integer-based date components (year, month, day of year, and day of week), and environmental spectral indices—specifically the Normalized Difference Vegetation Index (NDVI) as a vegetation indicator and the Normalized Difference Built-up Index (NDBI) as a built-up surface indicator. The primary modeling objective was set to estimate absolute Land Surface Temperature in degrees Celsius (lst_celsius). However, this raw dataset suffered from severe data gaps caused by frequent missing or invalid daily pixel readings due to cloud cover and satellite screening.

To address these data quality issues, the revised strategy keeps the target variable strictly fixed as absolute Land Surface Temperature (LST in °C) while refining the overall data structure. The environmental features continue to utilize NDVI and NDBI alongside fixed geographic coordinates across 116 grid points in Malang City. Rather than dropping incomplete records,

invalid satellite observations caused by cloud cover are now systematically resolved using location-wise mean imputation, where missing LST values are filled using historical averages computed for each specific spatial coordinate to ensure continuous spatio-temporal coverage.

Preprocessing

The initial preprocessing workflow relied on extracting raw date strings into discrete temporal integer components (year, month, day of year, and day of week) alongside basic boundary checks on spatial and environmental variables. However, treating date metrics as raw sequential integers introduced severe temporal artificial discontinuities—such as treating December 31st and January 1st as 365 days apart rather than consecutive days. As highlighted by Mahajan et al. (2021), raw integer temporal representations create boundary gaps that degrade regression models and prevent algorithms from learning smooth seasonal continuity.

To eliminate these pipeline bottlenecks, daily observations are aggregated into weekly temporal windows by calculating mean values for each location point. Temporal features are then encoded into continuous circular representations using trigonometric sine and cosine functions ($\sin$ / $\cos$) scaled across a 52-week annual cycle:

$$Week_{\sin} = \sin\left(\frac{2\pi \times Week}{52}\right), Week_{\cos} = \cos\left(\frac{2\pi \times Week}{52}\right)$$

Following the temporal encoding framework established by Khazem & Kanso (2025), this cyclical transformation eliminates boundary gaps between consecutive annual cycles and preserves natural seasonal periodicity. Furthermore, missing LST pixel values caused by cloud cover are imputed using location-specific historical means before model training.

Feature Extraction

The initial feature extraction strategy relied on an 8-predictor vector consisting of [longitude, latitude, year, month, day_of_year, day_of_week, ndvi, ndbi]. This formulation lacked interaction terms between environmental surface indicators and used raw integer encodings for temporal attributes, limiting the regression model's ability to learn complex spatial-temporal interactions.

To maximize predictive power, the revised feature vector incorporates spatial coordinates (longitude, latitude), environmental spectral indices (ndvi, ndbi), a composite interaction feature ($NDVI \times NDBI$) representing built-up land versus vegetation dynamics, and cyclical temporal features (week_sin, week_cos) derived from the 52-week annual cycle. The schema ordering is explicitly fixed within the FastAPI backend payload to ensure consistent data contracts during live model inference.

Training Data and Strategy

The primary candidate is the saved Train 2 artifact, urbanheat_ai_train2_candidate.joblib. It contains the model and metadata, including the expected feature order and target. The model is a Gradient Boosting Regressor. The artifact is loaded by the FastAPI backend; the environment used for inference pins scikit-learn to version 1.6.1 to match the serialized model.

The project has discussed time-aware evaluation cutoffs to reduce the risk of evaluating on data from the same temporal period used for training. Results must always be reported with the associated split/cutoff because metrics from different evaluation runs are not directly interchangeable.

Final train/test row counts and the exact split ratio are not available in the currently confirmed notes. They should be copied from the training script or evaluation output rather than inferred. If the model was evaluated using a chronological split, state the cutoff date and report the number of samples in each partition.

Evaluation run/split	Train/test samples	Status
Chronological split (cutoff 2026-08-21)	To be verified from training output	Reported metrics available
Candidate run (cutoff 2026-08-25)	To be verified from training output	Reported metrics available

For comparison, a separate training strategy was employed. This secondary pipeline utilizes a Linear Regression model trained exclusively on the mean Normalized Difference Vegetation Index (NDVI) to predict relative Land Surface Temperature anomalies (LST anomaly). The model was trained on a dataset covering 116 unique grid locations in Malang City with data from October 2025 to October 2026. The model was evaluated using a 5-fold spatial block cross-validation strategy with a block size of 0.03 degrees. This validation strategy was designed to assess spatial generalization to held-out blocks and reduce the risk of spatial data leakage.

Comparisons among the evaluated models showed that the single-feature Linear Regression model using NDVI achieved the best overall performance in the reported evaluation, outperforming the multi-feature Linear Regression and Random Forest configurations. The NDVI-only model achieved an R^2 of 0.528, an RMSE of 0.868, and an MAE of 0.649. These results suggest that NDVI alone provided a useful predictor of relative LST anomalies within the

evaluated dataset. However, the findings do not establish that NDVI is universally the best predictor or that the model can forecast future LST conditions.

To overcome the performance limits and negative validation scores of earlier raw setups, the updated training strategy incorporates feature preprocessing and statistical correlation screening based on Khazem & Kanso (2025). All engineered features, including weekly cyclical sine/cosine temporal encodings ($Week_{\sin}, Week_{\cos}$), spatial coordinates, environmental indices (NDVI, NDBI), and the composite interaction term ($NDVI \times NDBI$), undergo Pearson correlation (R) and R^2 screening against absolute LST before model fitting. Variables exhibiting near-zero linear correlation are eliminated. Candidate algorithms (Gradient Boosting, Random Forest, Linear Regression, and MLP Regressors) are being re-trained on the weekly aggregated and cyclically encoded dataset, with final model selection remaining open pending empirical evaluation.

Model/evaluation run	MAE	RMSE	R^2
Mean Baseline	1.072	1.349	-0.140
Linear Regression - Only NDVI	0.649	0.868	0.528
Linear Regression - All Features	0.661	0.883	0.512
Random Forest - All Features	0.781	1.032	0.333
MLP Regressor - Custom Master	1.2339	1.67569	0.6569

Model Building

The task is regression rather than classification because the output is a continuous temperature value in degrees Celsius (lst_celsius). The primary candidate uses Gradient Boosting Regression. The saved artifact is expected to include the model object, feature list, target name, model name, status, and dataset row count.

The inference backend is implemented with FastAPI. The health endpoint is GET /health, and the prediction endpoint is POST /api/predict. A prediction request contains latitude, longitude, observation_date, NDVI, and NDBI.

The backend derives the temporal features from observation_date and calls the trained model using the required feature order. To further optimize regression stability and eliminate boundary discontinuities, our modeling workflow incorporates advanced feature transformations, including cyclical time encoding ($\sin$ / $\cos$ representation for 52 weekly periods) as supported by Mahajan et al. (2021) and Khazem & Kanso (2025), alongside target scaling using StandardScaler.

Pipeline decision: the Train 2 model owned by the project lead is the primary LST-in-Celsius estimator. Secondary pipelines are treated as a complementary stage for spatial validation or hotspot-reference comparison. Its output should retain its original meaning and units; a relative LST anomaly must not be presented as an absolute temperature in Celsius. The exact interface and data contract between model stages are documented and enforced via the backend validation schema before live deployment.

Evaluation

Because this is a regression problem, evaluation should use regression metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2), rather than classification metrics such as accuracy, precision, recall, or F1-score.

Two Train 2 evaluation records are present in the project notes and must be kept separate because they use different evaluation cutoffs. For the run with a cutoff of 2026-08-21, the reported Train 2 metrics were MAE = 1.6423 °C, RMSE = 2.1923 °C, and R^2 = -0.1367. For a later Train 2 candidate run with a cutoff of 2026-08-25, the reported metrics were MAE = 1.4361 °C, RMSE = 1.8771 °C, and R^2 = 0.1951. These are preliminary results from separate runs, not a single combined result.

Model/evaluation run	MAE (°C)	RMSE (°C)	R^2
Train 2 - cutoff 2026-08-21	1. 6423	2. 1923	-0. 1367
Train 2 candidate - cutoff 2026-08-25	1. 4361	1. 8771	0.1951
MLP regressor (custom engine)	1.2339	1.6788	0.6569

Note: These rows represent separate evaluation runs and must not be merged into one result.

Linear Regression Model Using NDVI

The Linear Regression model using mean NDVI as its sole predictor achieved an MAE of 0.649, an RMSE of 0.868, and an out-of-fold R^2 of 0.528. Compared with the other tested configurations, the NDVI-only model achieved the lowest prediction errors and highest out-of-fold R^2. This suggests that NDVI alone was the most effective predictor of relative LST anomalies among the initial configurations evaluated. However, the model estimates relative LST anomalies rather than absolute land surface temperatures or future temperatures. Its performance on independent locations or later observations remains to be evaluated.

Post-Revision Evaluation Strategy

An R^2 value below or only modestly above zero indicates that the model's predictive performance is limited and requires further validation. The initial negative or low R^2 values occurred because raw integer date representations and unscaled coordinates lacked direct linear correlation with surface temperature.

To resolve these evaluation limitations, the pipeline was updated to incorporate cyclical time encoding ($Week_{\sin}, Week_{\cos}$), location-wise mean imputation for cloud-covered pixels, and target scaling as supported by Mahajan et al. (2021) and Khazem & Kanso (2025). This custom spatial-temporal feature engineering successfully mitigates circular date gaps and improves prediction stability across seasonal cycles. MAE and RMSE describe typical prediction error in Celsius, with RMSE penalizing larger errors more strongly. All candidate models must be evaluated against these standard regression metrics using reproducible chronological holdout splits.

Conclusion and Future Work

The UrbanHeat AI project has successfully identified key structural bottlenecks in its preliminary Land Surface Temperature (LST) regression pipeline. Initial evaluation runs demonstrated that relying on raw integer date components and unencoded spatial coordinates led to severe temporal discontinuities and poor or negative R^2 validation scores.

To build a scientifically grounded modeling framework, the project workflow was systematically upgraded. Aggregating daily satellite readings into weekly temporal windows, applying location-wise mean imputation for cloud-covered missing pixels, and encoding temporal attributes into continuous trigonometric sine and cosine functions ($Week_{\sin}, Week_{\cos}$) effectively eliminated boundary gaps between consecutive annual cycles. Supported by the findings of Mahajan et al. (2021) and Khazem & Kanso (2025), this spatial-temporal feature engineering strategy—combined with Pearson correlation (R) screening—provides a robust foundation for capturing non-linear seasonal dynamics and predicting continuous LST values in Lowokwaru District.

Task & Future Work

To transition the current machine learning pipeline into a fully functional and deployed application, the team has outlined the following immediate milestones:

- Backend API Integration: Securely connect the FastAPI inference backend endpoints (GET /health and POST /api/predict) to the Flutter application frontend to handle real-time spatial-temporal prediction payloads.
- Mobile Application Features: Implement interactive spatial heatmap visualizations and temperature alert notification systems within the Flutter user interface.
- Model Benchmark & Deployment: Complete empirical re-evaluation across all candidate regression algorithms (Gradient Boosting, Random Forest, Linear Regression, and

Multi-Layer Perceptron) on the newly encoded dataset, select the optimal model artifact based on MAE, RMSE, and R^2 metrics, and deploy the finalized model pipeline.

Team Member Contribution

Use the table below as a draft. Confirm each contribution with the team before submission; the role descriptions are based on the currently available project context.

Team Member Name	Contribution
Nasywa Qonita Ramadhani Han	Developed and compared the primary LST regression experiments in the urbanheat_ai_training workflow, including Train 1 as a location-and-time baseline and Train 2 with additional NDVI and NDBI environmental features. Evaluated the candidate models using regression metrics (MAE, RMSE, and R^2), selected Train 2 as the current primary candidate for absolute LST prediction in degrees Celsius, and documented the limitations of its preliminary evaluation results. Implemented the FastAPI inference backend to load the saved model artifact, validate request inputs, derive date-based features, and expose the /health and /api/predict endpoints. Tested the backend endpoints and planned safe integration with the existing Flutter frontend.
Muhammad Rizki	Collected and explored MODIS-derived LST datasets for Malang City, including daily and combined satellite observations. Developed a complementary Linear Regression workflow to analyze relative LST anomalies using mean NDVI (ndvi_mean) as the final predictor. Evaluated the model using five-fold spatial block cross-validation, achieving an MAE of 0.6494 °C, an RMSE of 0.8682 °C, and an R^2 of 0.5281. Also generated heat_predictions.json for 116 locations, assigning Low, Medium, and High hotspot categories to support complementary spatial analysis and hotspot identification.
Brian Serafino Donovan	Responsible for extracting and preprocessing the Normalized Difference Vegetation Index (NDVI) and calculating the Normalized Difference Built-up Index (NDBI). These environmental indicators are intended to serve as input features for the UrbanHeat AI land surface temperature prediction workflow. Also contributed to discussions on aligning the temporal availability of NDVI and NDBI composite data with daily LST observations.
Angelina Thennessa Jo	Integrated spatial coordinates and environmental spectral indices (NDVI and NDBI) into a master dataset. Developed, trained, and optimized an advanced Multi-Layer Perceptron (MLP) Regressor neural network pipeline featuring Cyclical Time Encoding (month_sin, month_cos, day_sin, day_cos) and Target Scaling

	(StandardScaler). This custom engine successfully mitigated circular date gaps and improved prediction stability, achieving an evaluated MAE of 1.2339 °C, an RMSE of 1.6788 °C, and a robust R^2 of 0.6569.
Bima Arya	Responsible for evaluating and comparing machine learning model performance using MAE, RMSE, and R^2 metrics. Analyzed evaluation results to identify model strengths, prediction errors, and limitations. Contributed to result visualization and technical documentation to support transparent interpretation of the model's predictive performance.